

Chapter: SQL Binary Data and Datatypes

What is Binary Data?

Everything inside a computer is ultimately stored as bits (0s and 1s).

Examples:

- Images (.jpg, .png)
- PDFs
- Audio (.mp3)
- Videos (.mp4)
- ZIP files
- Executable files (.exe)

These are binary files, not text.

For example, the first few bytes of a PNG image are:

```
89 50 4E 47 0D 0A 1A 0A
```

This is binary data, not readable text.

Character vs Binary Data

Suppose we store

```
Aadya
```

Internally (ASCII):

```
65 97 100 121 97
```

That's still text.

Now an image:

```
FF D8 FF E0 00 10 ...
```

Those bytes don't represent readable characters.

Hence SQL provides binary datatypes.

Binary Datatypes

Datatype	Purpose
BINARY	Fixed-length binary
VARBINARY	Variable-length binary
TINYBLOB	Small binary objects
BLOB	Medium binary objects
MEDIUMBLOB	Large binary objects
LOB	Very large binary objects

HashValue BINARY(16)

Suppose

```
ABC
```

Storage

```
ABC + remaining bytes padded with 0x00
```

Exactly like CHAR pads spaces,
BINARY pads with zero bytes.

VARBINARY

Like VARCHAR.

Stores variable-length binary data.

Syntax

```
Hash VARBINARY(64)
```

Now

5 bytes stores 5 bytes.

40 bytes stores 40 bytes.

Only the actual bytes plus a small length prefix are stored.

BLOB Family

BLOB

BLOB stands for **Binary Large Object**

Used for storing large binary data.

Syntax

```
Resume BLOB
```

Maximum Size

65,535 bytes ≈ 64 KB

Uses

- PDFs
- Small images
- Binary documents

TINYBLOB

Smallest BLOB.

Maximum

255 bytes

Uses

Tiny binary values.

Rarely used.

MEDIUMBLOB

Maximum

16,777,215 bytes ≈16 MB

Uses

- Images
- Medium videos
- Audio
- Documents

LOBLOB

Largest binary datatype.

Maximum

4,294,967,295 bytes ≈4 GB

Uses

- Movies
- Large videos
- Medical imaging
- Archives

Internal Storage

Large BLOB values are usually stored outside the main data row (especially in InnoDB), with the row containing a pointer to the external data.

This keeps rows reasonably compact while allowing storage of very large binary objects.

Advantages & Disadvantages

Advantages

- Stores any binary file.
- Supports very large objects.
- No manual encoding required.

Disadvantages

- Large storage.
- Slower backups.
- Higher replication cost.
- Larger indexes (if indexing prefixes).
- More memory and disk I/O.

Real Examples

Student Database

```
CREATE TABLE Student(  
    StudentID INT AUTO_INCREMENT PRIMARY KEY,  
    Name VARCHAR(100),  
    Email VARCHAR(255),  
    DOB DATE,  
    CGPA DECIMAL(3,2),  
    Gender ENUM('Male', 'Female', 'Other'),  
    Hosteller BOOLEAN  
);
```

Why?

Column	Reason
StudentID	INT is sufficient

Name	Variable length
Email	Variable length
DOB	Date only
CGPA	Exact decimal values
Gender	Limited choices
Hosteller	Yes/No

Master Decision Tree

Need Numbers?

— Integer?

- Small → TINYINT / SMALLINT
- Medium → INT
- Huge → BIGINT

— Decimal?

- Money → DECIMAL
- Scientific → DOUBLE

Need Text?

- Fixed Length → CHAR
- Variable Length → VARCHAR
- Large Text → TEXT/LONGTEXT

Need Date?

- Birthday → DATE
- Time Only → TIME
- Event → DATETIME
- Logs → TIMESTAMP

Need Binary?

- Fixed → BINARY
- Variable → VARBINARY
- Large Files → BLOB

|
Need Flexible Data?

|
└─ JSON

|
Need Fixed Choices?

|
└─ ENUM

|
└─ SET (MySQL only)